

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0549

Mickens 345 kV
December 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0549. The Interconnection Customer proposes a 262.4 MW solar photovoltaic facility interconnecting at the Mickens 345 kV substation in the Virginia area. The request was received on December 05, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 191 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,938,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Acec Lewiston-Hewitt Rd 138 kV	New Denver-Martin Dam 138 kV line	121.6	338	5.7
Daleville-New Enver 138 kV	New Denver-Greening 138 kV	105.2	479	7.0
Greening-Grahamsville 138 kV	Greening 345/138 kV XFMR	105.0	381	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hewitt Rd-Acec Lewiston 138 kV line	Rebuild line with higher-capacity conductor	\$35,154,000	7% (\$2,481,000)
Daleville-New Enver 138 kV	Reconductor line	\$7,480,000	4% (\$335,000)
Grahamsville-Greening 138 kV	Reconductor line	\$2,525,000	5% (\$122,000)
	Total allocated to project		\$2,938,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.